Computer Organization
Homework 4

1. Consider the following instruction:

   Instruction: addi Rt, Rs, 40

   Interpretation: Reg[Rt]=Reg[Rs] +40

   (a) What are the values of control signals generated by the control on page 24 of the slides for the above instruction? (8 control signals) (15%)

   (b) Which blocks perform a useful arithmetic function for this instruction? (5%)

   (c) Which blocks produce outputs, but their outputs are not used for this instruction? (5%)

   (d) Which blocks produce no outputs for this instruction? (5%)

2. Problems in this exercise assume that logic blocks needed to implement a processor's datapath have the following latencies:

I-Mem Read/ Write	Add	Mux	ALU	Regs Read/ Write	D-Mem Read/ Write	Sign-Extend	Sift-left-2
240ps	50ps	40ps	120ps	120ps	280ps	30ps	20ps

   (a) If the only thing we need to do in a processor is fetch consecutive instruction, what would the cycle time be? (Page 13 of the slides) (5%)

   (b) Consider a datapath similar to the one in Fig. 4.11 (Page 20 of the slides), but for a processor that only has R-type of instruction. What would the cycle time be for this datapath? (5%)

   (c) For the datapath element Shift left 2, which kinds of instructions require this resource block? What would the cycle time be for this kind of instructions? (5%)

   (d) Consider a datapath similar to the one in Fig. 4.11 (Page 20 of the slides), but for a processor that needs to support R type, load/save, branch instructions. What would the cycle time be for this datapath? (5%)

   (e) From (d), for which kinds of instructions (if any) is this resource on the critical path. (5%)

3. Problems in this exercise assume that logic blocks needed to implement a processor's datapath have the following latencies:

I-Mem	Add	Mux	ALU	Regs	D-Mem	Sign-Extend	Shift-Left-2
200ps	70ps	20ps	90ps	90ps	250ps	15ps	10ps

(a) If the only thing we need to do in a processor is fetch consecutive instructions , what would the cycle time be?(5%)

(b) Consider a datapath similar to the one in Figure 4.11, but for a processor that only has one type of instruction: unconditional PC-relative branch. What would the cycle time be for this datapath?(5%)

(c) Repeat (b), but this time we need to support only conditional PC-relative branches.(5%)

4. Problems in this exercise refer to the following sequence of instructions, and assume that it is executed on a 5-stage pipelined datapath:

```
add r5, r2, r1
lw   r3, 4 (r5)
lw   r2, 0 (r2)
or   r3, r5, r3
sw   r3, 0 (r5)
```

(a) If there is no forwarding or hazard detection, insert NOPs to ensure correct execution. (5%)

(b) Repeat 4.13.1 but now use NOPs only when a hazard cannot be avoided by changing or rearranging these instructions. You can assume register R7 can be used to hold temporary values in your modified code. (5%)

5. In this exercise, we examine how resource hazards, control hazards, and Instruction Set Architecture (ISA) design can affect pipelined execution. Problems in this exercise refer to the following fragment of MIPS code:

```
Sw    r16,12(r6)
Lw    r16 8(r6)
Beq   r5,r4,Label #Assume r5!=r4
Add   r5,r1,r4
Slt   r5,r15,r4
```

Assume that individual pipeline stages have the following latencies:

IF	ID	EX	MEM	WB
200ps	120ps	150ps	190ps	100ps

(a) For this problem, assume that all branches are perfectly predicted (this eliminates all control hazards) and that no delay slots are used. If we only

have one memory (for both instructions and data), there is a structural hazard every time we need to fetch an instruction in the same cycle in which another instruction accesses data. To guarantee forward progress, this hazard must always be resolved in favor of the instruction that accesses data What is the total execution time of this instruction sequence in the 5-stage pipeline that only has one memory?(5%)

(b) For this problem, assume that all branches are perfectly predicted (this eliminates all control hazards) and that no delay slots are used. If we change load/store instructions to use a register (without an offset) as the address, these instructions no longer need to use the ALU. As a result, MEM and EX stages can be overlapped and the pipeline has only 4 stages. Change this code to accommodate this changed ISA. Assuming this change does not affect clock cycle time, what speedup is achieved in this instruction sequence?(5%)

(c) Assuming stall-on-branch and no delay slots, what speedup is achieved on this code if branch outcomes are determined in the ID stage, relative to the execution where branch outcomes are determined in the EX stage?(5%)

(d) Given these pipeline stage latencies, repeat the speedup calculation from (c), taking into account the (possible) change in clock cycle time. Assume that the latency ID stage increases by 50% and the latency of the EX stage decreases by 10ps when branch outcome resolution is moved from EX to ID.(5%)

1. Consider the following instruction:
   Instruction: addi Rt, Rs, 40
   Interpretation: Reg[Rt]=Reg[Rs] +40
   (a) What are the values of control signals generated by the control on page 24 of the slides for the above instruction? (8 control signals) (15%)
   (b) Which blocks perform a useful arithmetic function for this instruction? (5%)
   (c) Which blocks produce outputs, but their outputs are not used for this instruction? (5%)
   (d) Which blocks produce no outputs for this instruction? (5%)

(a) RegDst = 0 # 使用 rt
Branch = 0
MemRead = 0
Mem to Reg = 0
ALUOp = 00 # 00 (加法) 01 (減法) 10 (根據 op code) R type
MemWrite = 0
ALUSrc = 1
RegWrite = 1

(b) ALU

(c) Add # Branch = 0

(d) Data Memory # MemRead = 0 , no output

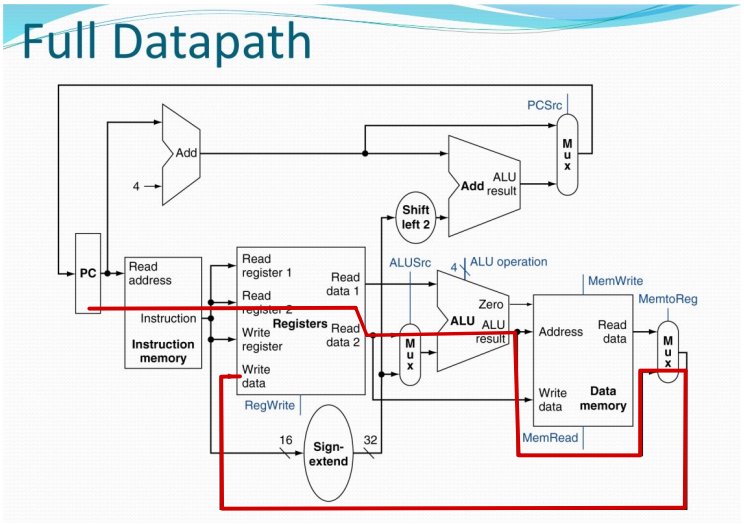
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I-Mem Read/ Write	Add	Mux	ALU	Regs Read/ Write	D-Mem Read/ Write	Sign-Extend	Sift-left-2
240ps	50ps	40ps	120ps	120ps	280ps	30ps	20ps

(a) If the only thing we need to do in a processor is fetch consecutive instruction, what would the cycle time be? (Page 13 of the slides) (5%)
(b) Consider a datapath similar to the one in Fig. 4.11 (Page 20 of the slides), but for a processor that only has R-type of instruction. What would the cycle time be for this datapath? (5%)
(c) For the datapath element Shift left 2, which kinds of instructions require this resource block? What would the cycle time be for this kind of instructions? (5%)
(d) Consider a datapath similar to the one in Fig. 4.11 (Page 20 of the slides), but for a processor that needs to support R type, load/save, branch instructions. What would the cycle time be for this datapath? (5%)
(e) From (d), for which kinds of instructions (if any) is this resource on the critical path. (5%)

(a) Instruction fetch → I-Mem read 240 ps #

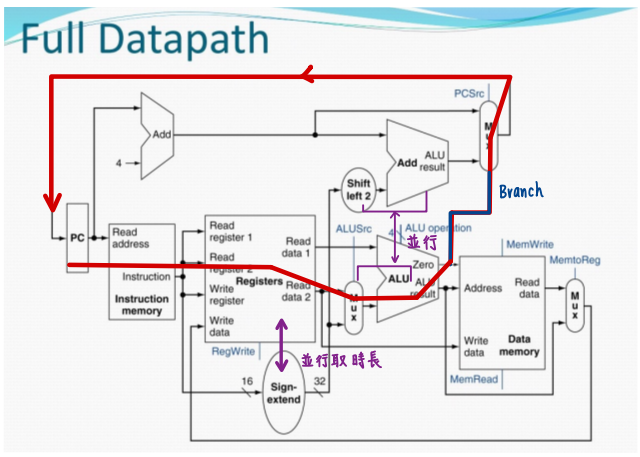
(b)



I-Mem read → Reg read → mux → ALU → mux → Reg write

240 + 120 + 40 + 120 + 40 + 120 = 680 ps #

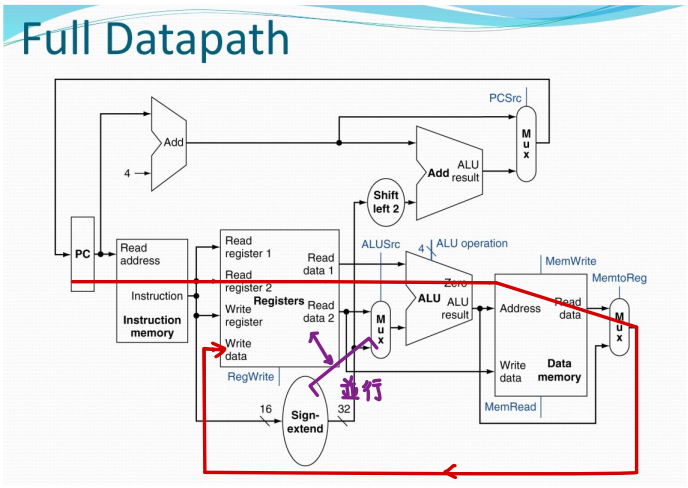
(c)



I-Mem read → Reg read → mux → ALU → mux

240 + 120 + 40 + 120 + 40 = 560 ps #

(d)



取指令時間最長做為 cycle time : load

I-Mem read → Reg read → ALU → D-Mem Read → mux → Reg write

240 + 120 + 120 + 280 + 40 + 120 = 920 ps #

(e) load #

3. Problems in this exercise assume that logic blocks needed to implement a processor's datapath have the following latencies:

I-Mem	Add	Mux	ALU	Regs	D-Mem	Sign-Extend	Shift-Left-2
200ps	70ps	20ps	90ps	90ps	250ps	15ps	10ps

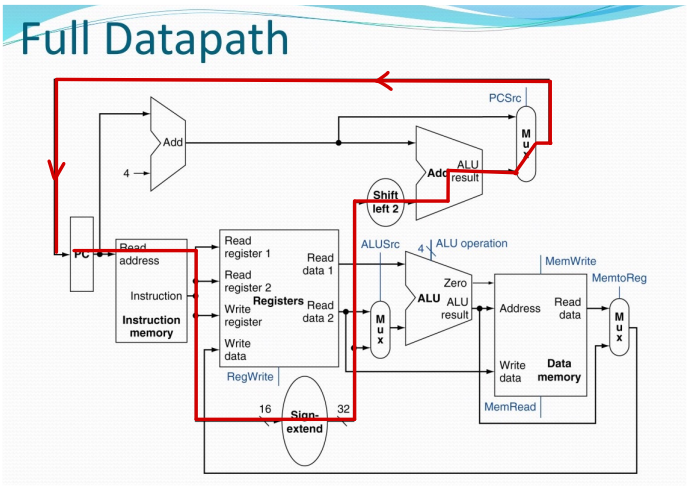
(a) If the only thing we need to do in a processor is fetch consecutive instructions , what would the cycle time be?(5%)

(b) Consider a datapath similar to the one in Figure 4.11, but for a processor that only has one type of instruction: unconditional PC-relative branch. What would the cycle time be for this datapath?(5%)

(c) Repeat (b), but this time we need to support only conditional PC-relative branches.(5%)

(a) I - Mem : 200 ps #

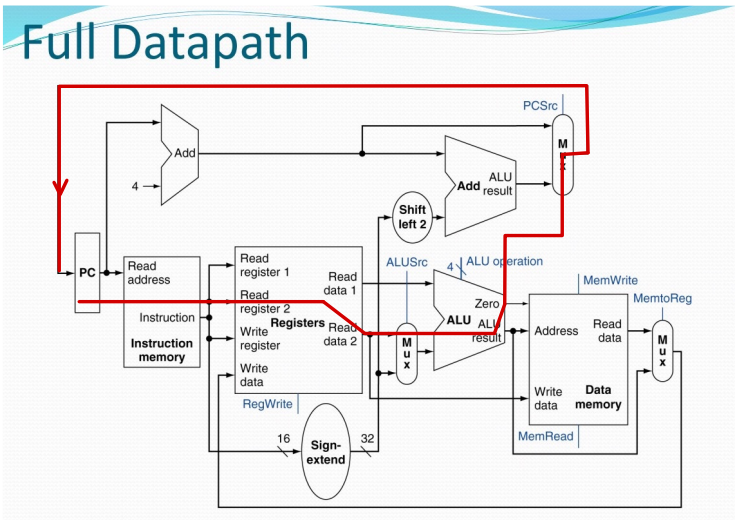
(b)



I-Mem read → sign extend → shift left 2 → Add → mux

200 + 15 + 10 + 70 + 20 = 315 ps #

(c)



I-Mem read → Reg read → mux → ALU → mux

200 + 90 + 20 + 90 + 20 = 420 ps #

4. Problems in this exercise refer to the following sequence of instructions, and assume that it is executed on a 5-stage pipelined datapath:

```
add r5, r2, r1
lw  r3, 4 (r5)
lw  r2, 0 (r2)
or  r3, r5, r3
sw  r3, 0 (r5)
```

(a) If there is no forwarding or hazard detection, insert NOPs to ensure correct execution. (5%)

(b) Repeat 4.13.1 but now use NOPs only when a hazard cannot be avoided by changing or rearranging these instructions. You can assume register R7 can be used to hold temporary values in your modified code. (5%)

(a)

```
add r5, r2, r1
nop
nop
lw  r3, 4 (r5)
lw  r2, 0 (r2)
nop
or  r3, r5, r3
nop
nop
sw  r3, 0, (r5)
```

WB 前半周期寫回，ID 後半周讀出

```
F  D  E  M  W
   *  *  F  D  E  M  W
            F  D  E  M  W
               *  F  D  E  M  W
                  *  *  F  D
```

* 同顏色表示 data hazard

(b)

```
add r5, r2, r1
lw  r2, 0 (r2)
nop
lw  r3, 4 (r5)
nop
nop
or  r7, r5, r3
sw  r7, 0 (r5)
```


add r5, r2, r1
lw r3, 4 (r5)
lw r2, 0 (r2)
or r3, r5, r3
sw r3, 0 (r5)

5.

(a)

```
1 2 3 4 5 6 7 8 9 10 11
F D E M W
  F D E M W
    F D E M W
      * * F D E M W
            F D E M W
```

∵ only one memory

∴ IF、Mem 不能同 clk 動作

cycle time = 200ps (IF) longest

200 × 11 = 2200 ps #

(b)

Old

```
1 2 3 4 5 6 7 8 9
F D E M W
  F D E M W
    F D E M W
      F D E M W
        F D E M W
```

New

```
1 2 3 4 5 6 7 8
F D M W
  F D M W
    F D M W
      F D M W
        F D M W
```

$\frac{9}{8} = 1.125$ #

(C)

* 成立

beq
add
slt
Label

```
F D E M W
  F D E M W
    F D E M W
      F D E M W
```

假設branch outcome在EXE判定，成立時跳到label，可以發現下面已經讀取2個指令，因此要2個stall來flush掉

* MEM 3 stall
EXE 2 stall
ID 1 stall

```
1 2 3 4 5 6 7 8 9
F D E M W
  F D E M W
    F D E M W
      F D E M W
        F D E M W
```

EXE = 9 + 2 = 11

ID = 9 + 1 = 10

$\frac{11}{10}$ = 1.1 times faster #

(d)

Cycle time還是以IF 200ps最長，有改變的如上一題一樣

$\frac{11}{10}$ = 1.1 times faster #